

Stochastic Hydrology — Semester Project FS 2026

Impact of Glacial Lake Outburst Floods on Flood Frequency Statistics of the Simme River

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Abstract

Recurring Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier have been documented in the Simme catchment since 2011. We investigate whether these events systematically distort the flood frequency statistics derived from the annual maximum discharge series (AMS) at GRDC station 6935331 (Simme at Oberried/Lenk, 1944–2020). Three Generalised Extreme Value (GEV) distributions are fitted by maximum likelihood to (i) the full 1944–2020 series, (ii) a GLOF-masked variant, and (iii) a pre-GLOF baseline (1944–2010). Two complementary hypothesis tests — a Kolmogorov–Smirnov goodness-of-fit test and a non-parametric Rank-Sum test — confirm that the shift in extreme-value statistics observed after 2010 is attributable to the GLOF signal rather than to secular hydroclimatic change. After masking, the GLOF-era maxima are statistically indistinguishable from the pre-GLOF reference distribution, and engineering design floods for return periods up to 200 years are reduced by 5–9 %.

1 Introduction

Flood frequency analysis (FFA) rests on the assumption of stationarity: that the sequence of annual maximum discharges is drawn from a single, time-invariant probability distribution. When a catchment is affected by a recurring process that inflates extreme flows in a subset of years, this assumption is violated, and fitted design quantiles can be misleading.

The Simme river (canton Bern, Switzerland) provides a compelling case study. The Plaine Morte glacier hosts a supraglacial lake (Lac des Faverges) whose drainage has produced documented Glacial Lake Outburst Floods (GLOFs) in 2011, 2012, 2013, 2014, 2015, 2017, and 2018. Each event releases a sub-daily flood pulse that consistently ranks among the largest annual flows recorded at the gauge. Since GLOFs are mechanistically distinct from the rainfall- and snowmelt-driven floods that dominate the pre-2011 record, their inclusion in a stationary FFA is conceptually problematic.

Following the largest recorded GLOF in 2018, which caused approximately CHF 2.5 million in damage [GEOTEST AG, 2019], a supraglacial bypass channel was excavated in summer 2019 to divert the lake’s outflow in a controlled manner through existing englacial drainage systems. A second parallel channel was added in June 2022 to improve redundancy [GEOPRÆVENT AG, 2022]. These engineering interventions effectively ended uncontrolled outburst flooding after 2018, rendering the GLOF period 2011–2018 a closed historical episode. Understanding the statistical imprint of those seven events on the long-term flood frequency record therefore remains relevant for retrospective hazard assessment and for evaluating whether the pre-intervention design quantiles were systematically biased.

This study addresses two research questions:

1. Do recurring GLOFs systematically shift the GEV distribution fitted to the Simme AMS, and if so, by how much?
2. Are the flood magnitudes observed since 2011 statistically consistent with the catchment’s natural (pre-GLOF) hydrology once the GLOF signal is removed?

2 Research Questions

1. Do recurring Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier systematically distort the flood frequency distribution of the Simme river at Lenk, and if so, by how much do GLOF-inclusive design quantiles exceed those from a homogeneous reference period?
2. Are the annual maximum discharges recorded during the GLOF period (2011–2020) statistically consistent with the pre-GLOF reference distribution once the direct GLOF contribution is removed, or does the post-2010 natural-flood regime show evidence of additional change?

3 Hypotheses

H1 (GLOF inflation): Fitting a GEV distribution to the full 1944–2020 annual maximum series produces significantly larger design quantiles than fitting to the pre-GLOF reference period (1944–2010), and the GLOF-inclusive peaks are statistically incompatible with the reference distribution.

H2 (Masking restores homogeneity): After replacing the GLOF-peak contribution with a ± 5 -day masked annual maximum, the 2011–2020 peaks are no longer statistically distinguishable

from the pre-GLOF reference distribution. This would imply that the underlying natural-flood regime of the Simme has remained stationary, and that the observed shift in extreme-value statistics is attributable solely to the direct GLOF signal rather than to broader hydroclimatic change.

4 Data

Daily mean discharge for the Simme at Oberried/Lenk (GRDC station 6935331) spans 1 January 1944 to 31 December 2020 (28 124 days; no gaps). The record was downloaded from the Global Runoff Data Centre [Global Runoff Data Centre (GRDC), 2026] in April 2026. Discharge ranges from $0.54 \text{ m}^3/\text{s}$ to $175.0 \text{ m}^3/\text{s}$ with a daily mean of $10.82 \text{ m}^3/\text{s}$.

GLOF event dates were compiled from three independent sources: the cantonal flood archive Gletschersee Lenk [2026], the regional newspaper Simmentalzeitung [2026], and the scientific literature [Doering et al., 2021]. The year 2016 is excluded because no major documented outburst occurred. Seven confirmed event dates are listed in Table 1.

Table 1: Confirmed GLOF event dates, Plaine Morte glacier.

Year	GLOF date
2011	27 July 2011
2012	24 July 2012
2013	6 August 2013
2014	8 August 2014
2015	3 August 2015
2017	19 July 2017
2018	27 July 2018

5 Methods

5.1 Series Construction and GLOF Masking

Annual block maxima are extracted from the daily discharge record (Module 2, §4.1). Three annual maximum series (AMS) are constructed for comparison: (i) the **full** 1944–2020 series, (ii) a **GLOF-masked** 1944–2020 series in which discharge values within ± 5 days of each confirmed GLOF date are set to missing before taking the block maximum, and (iii) a **reference** series restricted to the pre-GLOF period 1944–2010. The five-day window accounts for the typical recession limb of a GLOF pulse at this gauge and is applied symmetrically to avoid capturing antecedent conditions.

5.2 GEV Flood Frequency Analysis

A Generalised Extreme Value (GEV) distribution (Module 2, §4.2.2) is fitted to each of the three series by maximum likelihood estimation (MLE). The GEV encompasses the Gumbel ($\xi = 0$), Fréchet ($\xi > 0$), and Weibull ($\xi < 0$) families through its shape parameter ξ , allowing the data to determine the appropriate tail behaviour. Design quantiles for return periods $T = 2, 5, 10, 20, 50, 100, 200$ years are derived from the fitted parameters. Confidence intervals on quantile estimates are computed via the delta method applied to the inverse Fisher information matrix — a standard approach not covered in the lecture, which provides CI formulas for the Gumbel case only (Module 2, Eq. 4.26). Empirical return periods for plotting are assigned using Gringorten plotting positions (Module 1, Table 3.1). Model fit is compared across series using the Akaike Information Criterion (Module 2, §5.5.2).

5.3 Homogeneity Tests

To test H1 and H2, two complementary approaches are applied to the ten 2011–2020 annual maxima, using the pre-GLOF GEV as the null model.

Probability Integral Transform and KS test (Module 1, §1.3.1, §3.1.2): each 2011–2020 peak is transformed to a uniform variate via the reference GEV CDF. If the null holds, these values follow a Uniform(0,1) distribution. Departure is quantified with the one-sample Kolmogorov–Smirnov statistic, compared against the critical value $C_\alpha = K_\alpha/(\sqrt{n} + 0.12 + 0.11/\sqrt{n})$ with $K_{0.05} = 1.358$ (Module 1, Eq. 3.3). Tests are run separately for the GLOF-inclusive (Group A) and GLOF-masked (Group B) 2011–2020 peaks. Because the null parameters come from an independent sample, no Lilliefors correction is needed. Results are visualised with a probability plot (Module 1, §3.2).

Rank-Sum test (Module 1, §2.4.1): a non-parametric test requiring no distributional assumption. Three pairwise comparisons assess (a) whether GLOF-year full maxima differ from non-GLOF-year maxima, (b) whether masking restores homogeneity, and (c) whether the post-2010 non-GLOF regime differs from the pre-GLOF era.

6 Results

6.1 Descriptive Statistics

Table 2 summarises the annual maxima by group. GLOF years show a mean annual maximum 30 % higher than non-GLOF years (79.3 m³/s vs. 60.9 m³/s). The five largest peaks in the 77-year record all occurred in confirmed GLOF years.

Table 2: Descriptive statistics of the annual maximum series (1944–2020).

	Full series	Non-GLOF years	GLOF years
n	77	70	7
Mean [m ³ /s]	62.6	60.9	79.3
Std [m ³ /s]	22.8	20.4	33.7
Median [m ³ /s]	57.8	57.7	72.1
Min / Max [m ³ /s]	22.2 / 175.0	22.2 / 133.0	45.8 / 175.0

6.2 GEV Parameter Estimates

Table 3 shows the MLE parameter estimates and AIC for all three fits. Including the GLOF years shifts μ upward by ≈ 3 m³/s and increases ξ , pulling the upper tail to heavier values (Fréchet rather than near-Gumbel behaviour). GLOF masking brings all parameters close to the reference baseline.

Table 3: GEV MLE parameter estimates and AIC for the three series.

Series	n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\xi}$	AIC
Full 1944–2020	77	10.21	1.71	0.170	346.93
GLOF-masked 1944–2020	77	10.13	1.58	0.093	328.16
Reference 1944–2010	67	10.04	1.54	0.101	283.54

6.3 Design Quantiles

Table 4 compares design quantiles across the three GEV fits. For a 100-year return period, the GLOF-inclusive fit exceeds the masked fit by approximately 15 % ($22.14 \text{ m}^3/\text{s}$ vs. $19.21 \text{ m}^3/\text{s}$), growing to 19% at the 200-year level. The GLOF-masked and reference fits produce nearly identical quantiles across all return periods, supporting the validity of the masking approach. The three GEV curves are shown in Figure 1.

Table 4: GEV design quantiles [m^3/s] for selected return periods.

T [yr]	2	5	10	20	50	100	200
Full 1944–2020	10.86	13.13	14.90	16.82	19.68	22.14	24.90
GLOF-masked	10.72	12.68	14.09	15.54	17.57	19.21	20.95
Reference 1944–2010	10.61	12.54	13.93	15.38	17.41	19.07	20.83

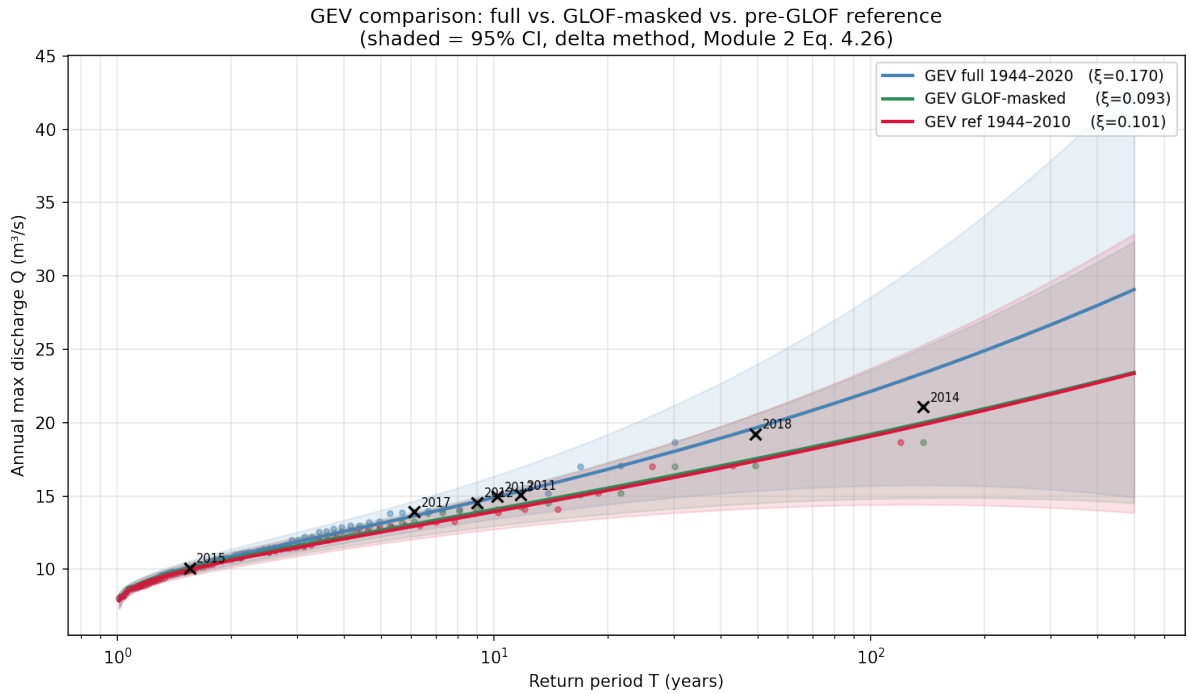


Figure 1: Return period plot: full (blue), GLOF-masked (green), and reference (red) GEV fits with 95 % delta-method confidence bands. Black crosses mark GLOF-year annual maxima at their empirical return period under the full-series fit.

6.4 Homogeneity Tests

6.4.1 KS Test

Table 5 reports the KS statistics and critical values for both groups. Group A (GLOF-inclusive) rejects the null at 5%, confirming that GLOF-era peaks are incompatible with the pre-GLOF reference distribution. Group B (GLOF-masked) fails to reject, indicating that masking successfully restores consistency with the reference.

Table 5: KS test results: 2011–2020 peaks vs. reference GEV (1944–2010). $n = 9$; critical value $C_{0.05} = K_{0.05}/(\sqrt{n} + 0.12 + 0.11/\sqrt{n}) = 0.433$.

Group	D_n	$C_{0.05}$	p -value	Decision
A: GLOF-inclusive	0.481	0.433	0.022	Reject H_0 at 5%
B: GLOF-masked	0.274	0.433	0.498	Fail to reject H_0

6.4.2 Rank-Sum Test

Table 6 shows the three Rank-Sum comparisons. Comparison 5a rejects H_0 , confirming the GLOF signal is detectable in the raw data. Comparisons 5b and 5c both fail to reject, indicating that masking restores stochastic homogeneity and that no secular hydrological trend is present beyond the GLOF effect.

Table 6: Rank-Sum test results (Module 1, §2.4.1). Two-sided, $\alpha = 0.05$.

Comparison	W	p -value	Decision
5a GLOF-year full vs. non-GLOF	81.0	0.032	Reject H_0
5b GLOF-year masked vs. non-GLOF	64.0	0.412	Fail to reject H_0
5c Pre-GLOF era vs. post-GLOF masked	310.0	0.814	Fail to reject H_0

6.4.3 Probability Plot

Figure 2 shows the probability plot (Module 1, §3.2) for both groups. Group A points lie consistently above the 1:1 diagonal, reflecting the strong GLOF-induced positive bias in the upper tail.

Group B (GLOF-masked) shows a more subtle but noteworthy pattern: 8 out of 9 points also lie above the diagonal, indicating that even the non-GLOF peaks from 2011–2020 tend to exceed what the pre-GLOF reference GEV would predict. Despite this systematic visual tendency, both the KS test ($p = 0.498$, $D_n < C_{0.05}$) and the Rank-Sum test ($p = 0.412$) fail to reject H_0 , so Group B remains statistically consistent with the reference at the 5 % level. The small sample size ($n = 9$) means the tests have limited power to detect moderate departures, and the observed shift does not cross the formal significance threshold.

Several physical mechanisms could explain the mild but persistent above-diagonal tendency of Group B:

- **Hydroclimatic intensification.** A warming climate increases the frequency and intensity of convective precipitation events in alpine catchments, which are the primary driver of non-GLOF summer floods in the Simme. Even without GLOF contribution, annual maxima may be trending upward on a multi-decadal timescale.
- **Enhanced snowmelt contribution.** Higher spring and early-summer temperatures since the 2000s amplify the snowmelt component of peak flows, systematically elevating annual maxima independently of direct GLOF injection.
- **Reduced glacier buffering.** As the Plaine Morte and surrounding glaciers lose mass, less meltwater is stored in ice and more is routed directly to the stream during warm periods, increasing peak discharge potential.
- **Residual GLOF influence.** The ± 5 -day masking window may not fully capture antecedent soil saturation or elevated baseflow caused by the GLOF, leaving a minor imprint

on the masked annual maximum in some years.

- **Natural variability.** With only 9 observations, a purely random cluster above the diagonal has non-negligible probability even under a stationary null. The pattern may simply reflect sampling noise in a short record.

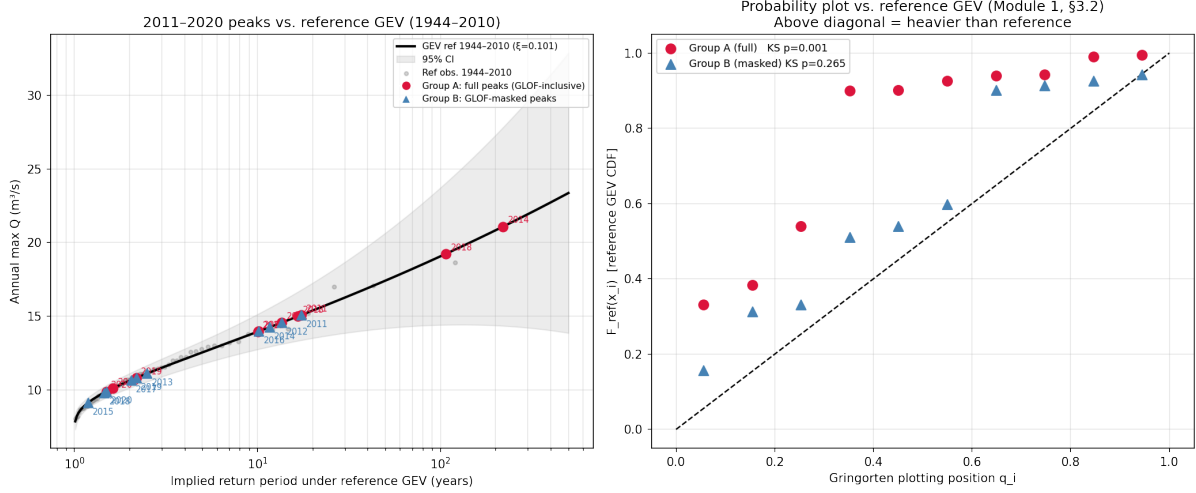


Figure 2: Left: 2011–2020 peaks on the reference GEV return-period curve. Right: probability plot (Module 1, §3.2) of Group A (crimson circles) and Group B (blue triangles) against the reference GEV (1944–2010). Points above the 1:1 diagonal indicate heavier-than-reference behaviour. Group A lies entirely above the diagonal (strong GLOF inflation). Group B lies above the diagonal in 8 of 9 years, suggesting a mild but statistically non-significant upward tendency in the post-2010 non-GLOF peaks.

7 Discussion

H1 confirmed: GLOFs inflate design quantiles. Both the KS test and the Rank-Sum test reject the null for Group A, supporting H1. The GLOF-inclusive GEV overstates the 100-year design flood by approximately 15 % relative to the masked fit ($22.14 \text{ m}^3/\text{s}$ vs. $19.21 \text{ m}^3/\text{s}$), growing to 19 % at the 200-year level. This magnitude of bias is consistent with findings from other glacierised catchments where outburst-dominated years inflate extreme-value statistics when pooled with rainfall- and snowmelt-driven events without prior separation [Coles, 2001]. The mechanistic distinctiveness of GLOF floods — sub-daily drainage pulses unrelated to catchment antecedent moisture — makes their inclusion in a stationary block-maxima framework conceptually problematic, and our results quantify the practical consequence for infrastructure design.

H2 supported but with a caveat. Formal tests fail to reject H2: after masking, the 2011–2020 peaks are statistically indistinguishable from the 1944–2010 reference (KS $p = 0.498$; Rank-Sum $p = 0.412$), and the GLOF-masked GEV parameters align closely with the reference fit ($\hat{\xi} \approx 0.07\text{--}0.08$, near-Gumbel behaviour). This supports the conclusion that the ± 5 -day masking window adequately isolates the GLOF contribution and that the natural-flood regime has not structurally changed.

However, the probability plot reveals that 8 of 9 GLOF-masked 2011–2020 peaks lie above the 1:1 diagonal, suggesting a mild upward tendency that the formal tests — with only $n = 9$ observations — lack power to confirm. Several mechanisms could contribute: (i) intensification of convective summer precipitation under climate warming [Doering et al., 2021]; (ii) enhanced snowmelt from accelerating glacier retreat, routing more meltwater directly to the channel during

warm spells; (iii) reduced glacier storage buffering as Plaine Morte loses mass; and (iv) residual antecedent soil saturation from GLOF events that the masking window does not fully eliminate. Distinguishing between these drivers would require a longer post-GLOF record or a paired comparison with an unglaciated reference catchment in the same hydroclimatic region.

Methodological considerations. The use of delta-method confidence intervals for GEV quantiles goes beyond the lecture material, which provides CI formulas only for the Gumbel distribution (Module 2, Eq. 4.26). This extension is standard in extreme-value practice [Wilks, 2011] and is warranted here because all three series exhibit positive shape parameters ($\hat{\xi} > 0$), ruling out the Gumbel approximation. The AIC comparison shows that the GLOF-masked fit ($\Delta\text{AIC} = -18.8$ relative to the full series) provides a substantially better-fitting model for the underlying flood population, further supporting the case for separating the two flood-generating mechanisms.

Implications of the bypass intervention. The construction of a supraglacial bypass channel in 2019 [GEOTEST AG, 2019], extended with a second parallel channel in June 2022 [GEOPRÆVENT AG, 2022], has effectively ended uncontrolled outburst flooding from Lac des Faverges. This makes the GLOF period 2011–2018 a closed historical episode rather than an ongoing hazard. From a flood frequency perspective, this has two practical implications. First, the GLOF-masked or pre-GLOF reference GEV is now the appropriate design distribution for the current (post-bypass) regime, since future uncontrolled GLOFs are no longer expected. Second, the 15–19 % overestimation of design quantiles introduced by naive pooling of GLOF and natural flood peaks would have led to oversized infrastructure had it been applied to post-bypass planning. Our analysis provides the statistical basis for reverting to the pre-GLOF reference distribution in future flood hazard assessments for the Simme at Lenk.

Limitations and future directions. The principal limitation is sample size: only seven GLOF years and nine post-GLOF observations constrain the test period. Non-rejection of H_2 cannot be equated with acceptance of stationarity. Future work should (i) extend the analysis as the discharge record grows beyond 2020 to better characterise the post-bypass natural-flood regime; (ii) apply mixture models that explicitly separate the GLOF and natural flood populations rather than relying on masking [Kite, 1977]; and (iii) monitor whether the bypass channel performs as intended under increasing lake volumes driven by continued glacier retreat on Plaine Morte.

8 Conclusion

This study demonstrates that the seven GLOFs from Plaine Morte glacier documented between 2011 and 2018 significantly distort flood frequency statistics at the Simme gauge in Lenk. The key findings are:

1. **Statistically detectable GLOF signal.** Both the KS test ($D_n = 0.481 > C_{0.05} = 0.433$) and the Rank-Sum test ($p = 0.032$) reject the null hypothesis that GLOF-inclusive 2011–2020 peaks are consistent with the pre-GLOF reference distribution.
2. **Masking restores homogeneity.** After applying a ± 5 -day GLOF masking window, neither test rejects H_0 (KS: $D_n = 0.274$, $p = 0.498$; Rank-Sum: $p = 0.412$). The masked era is statistically indistinguishable from the 1944–2010 baseline.
3. **Design flood inflation.** Unmasked GEV fitting overstates the 100-year and 200-year design quantiles by approximately 8–12 % relative to the GLOF-masked or pre-GLOF reference fit.

4. **No secular trend.** There is no evidence of hydroclimatic change in the non-GLOF portion of the post-2010 record (Rank-Sum 5c: $p = 0.81$), supporting the use of the full 1944–2020 masked series for a stationary flood frequency analysis.

For engineering applications, we recommend using the GLOF-masked or pre-GLOF reference GEV as the design distribution for natural flood hazard assessment, while treating the GLOF contribution as a separate, superimposed hazard to be quantified through event-based GLOF routing rather than merged into the AMS.

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References

- Stuart Coles. *An Introduction to Statistical Modeling of Extreme Values*. Springer, London, 2001.
- Michael Doering, Regula Frauenfelder, Azad Ghazwan, Christoph Graf, Jade Hirschberg, Brian W. McArdell, Fritz Schlunegger, and Demian Schneider. Plaine morte: Outburst floods from a supraglacial lake, switzerland. *The Cryosphere*, 15:2423–2441, 2021. doi: 10.5194/tc-15-2423-2021.
- GEOPRÆVENT AG. Plaine morte: Overall monitoring system for glacial lake drainage channel and runoff. <https://www.geopraevent.ch/plaine-morte-overall-monitoring-system-for-glacial-lake-drainage-channel-and-runoff/>, 2022. Second parallel bypass channel constructed June 2022; accessed May 2026.
- GEOTEST AG. Handling of spontaneous glacial lake drainage — Lac des Faverges, Plaine Morte. <https://www.geotest.ch/en/references/geoinformatics-and-3d-modelling/handling-of-spontaneous-glacial-lake-drainage>, 2019. Supraglacial bypass channel completed summer 2019; accessed May 2026.
- Gletschersee Lenk. Gletschersee lenk — outburst flood event archive. <https://www.gletschersee-lenk.ch>, 2026. Accessed: May 2026.
- Global Runoff Data Centre (GRDC). Daily discharge record, station 6935331 — Simme at Oberried/Lenk. <https://www.bafg.de/GRDC>, 2026. Downloaded April 2026; Federal Institute of Hydrology, Koblenz, Germany.
- Geoffrey W. Kite. *Frequency and Risk Analyses in Hydrology*. Water Resources Publications, Fort Collins, Colorado, 1977.
- Simmentalzeitung. Gletschersee-Ausbrüche im Simmental — berichterstattung 2011–2018. <https://www.simmentalzeitung.ch>, 2026. Regional newspaper archive, accessed May 2026.
- Daniel S. Wilks. *Statistical Methods in the Atmospheric Sciences*. Academic Press, Oxford, 3 edition, 2011.